Teaching and Lesson Design Handbook

Lesson planning, sequencing, examples, practice, pacing, and review.

Reference material for lesson planning and course-content development.

Contents

1. Planning from the learning goal

Start with what learners should be able to do by the end of the lesson. A topic label such as 'Python loops' is too broad on its own. A useful objective states an observable outcome, for example: 'Use a for-loop with range() to repeat a short block of code' or 'Explain why a while-loop stops'.

A 60-minute lesson should not try to cover an entire subject. Choose a small set of outcomes that fit the available time. If the lesson has six objectives and every objective needs a new concept, the scope is probably too large.

Useful objective verbs

- Explain
- Identify
- Compare
- Trace
- Calculate
- Use
- Debug
- Create

Avoid vague objectives such as 'understand loops' unless they are followed by specific evidence of understanding. Objectives should guide examples, practice, and assessment. If an assessment item does not connect to an objective, ask why it is included.

2. Sequencing a lesson

A clear sequence lowers unnecessary cognitive load. Learners should not have to understand three new ideas at once just to see the main idea. Start from a familiar problem, introduce one concept, show it in action, then give learners a chance to use it.

A practical sequence

- Short opening or diagnostic question.
- Direct explanation of the first idea.
- Worked example.
- Guided practice.
- Second idea or extension if needed.
- Independent practice.
- Quick check and recap.

This sequence is not a rigid template. A science activity may place observation before explanation. A communication lesson may begin with two sample emails and ask learners to compare them. The point is that the sequence should make sense for the learning goal.

3. Worked examples that actually teach

A worked example is more than a finished answer. It should show the decisions and intermediate steps that a learner would otherwise have to guess. For programming, this may be a trace of variables. For mathematics, it may be the reason for each algebraic step. For science, it may be a diagram with labeled quantities.

Good worked-example habits

- Keep the first example small.
- Use descriptive variable names or labels.
- Show one important intermediate state.
- Explain why the chosen step is appropriate.
- End by checking the result.

Later examples can remove some support. This gradual fading helps learners move from watching to doing. Avoid making every example so detailed that learners never have to make a decision themselves.

4. Using misconceptions as teaching material

Many topics have predictable wrong ideas. These are useful because they show where a lesson can go beyond definitions. A Python learner may think range(1, 5) includes 5. A mathematics learner may 'move a term across the equals sign' without understanding the operation. A science learner may think a moving object always needs a forward net force.

A good lesson can include a short incorrect example and ask learners to diagnose it. The wrong example should be realistic. Avoid cartoonishly bad errors that nobody would make.

Misconception routine

- Show the claim or solution.
- Ask whether it is correct.
- Ask for the specific reason.
- Correct the reasoning, not only the final answer.
- Give a second example to confirm the correction.

The same approach works in communication lessons. Learners can compare an unclear email with a clearer version and identify what changed.

5. Guided practice and independent practice

Guided practice gives learners a chance to use the new idea while support is still available. Independent practice checks whether they can act without the teacher providing every next step. The two should not be identical.

Guided practice

Provide prompts, partial steps, or a structured table. Ask questions such as 'what changes next?' or 'which condition are we checking?'.

Independent practice

Remove some prompts and change the context. The learner should still recognize the same underlying idea. If the context is too different, the exercise may test unrelated knowledge instead.

For a 60-minute lesson, three well-chosen practice tasks can be more useful than fifteen repetitive ones. Include at least one task that exposes reasoning, not just a final answer.

6. Adapting for beginner and intermediate learners

A beginner version should reduce assumed prior knowledge and make intermediate steps visible. It should not simply remove half the content. An intermediate version can combine ideas that were already taught, reduce hints, and use less familiar contexts.

Beginner adjustments

- Define new terms when first used.
- Use smaller examples.
- Show traces or intermediate steps.
- Keep instructions short.
- Include hints that point to a next step.

Intermediate adjustments

- Combine two previously taught ideas.
- Ask learners to choose between methods.
- Use debugging or comparison tasks.
- Reduce step-by-step hints.
- Require a short explanation of the choice.

An advanced variant should preserve the same core objective when it is intended as a variant of one lesson. Introducing a completely new topic creates a different lesson, not a harder version.

7. Pacing and lesson length

The stated duration should affect the amount of content. A 45-minute lesson needs fewer examples and less independent practice than a 90-minute workshop. The lesson pack should not generate the same amount of material regardless of duration.

Duration	Typical shape
30-45 minutes	One main concept, one worked example, short practice, quick check
60 minutes	Two related ideas, two worked examples, guided and independent practice, short quiz
75-90 minutes	More practice, one extension, longer activity, fuller review

Build in a small amount of flexibility. Some classes need more time on the first example. A lesson plan that uses every minute with no buffer is difficult to teach in practice.

If a topic cannot fit the requested duration without rushing, split it into a sequence rather than compressing every section.

8. Language and readability

Use plain language where possible. Technical terms should be used when they matter, but they should be defined. Avoid replacing a simple sentence with formal wording that does not add meaning.

Readable structure

- Use descriptive headings.
- Keep paragraphs focused on one idea.
- Separate examples from explanation.
- Use numbered steps for procedures.
- Keep code and formulas copyable as text.

Do not rely on formatting alone to communicate meaning. A learner reading plain text should still understand which answer is correct, which step comes next, or which value belongs to which label.

9. Activities and labs

An activity should make learners do something observable. Reading a paragraph and answering a recall question is not much of a lab. Better activities ask learners to modify, measure, compare, classify, debug, or produce something.

Computing

Change a parameter and compare output, trace a program, fix a bug, or compare two implementations.

Mathematics

Create a table, graph a relationship, compare strategies, or analyze an incorrect solution.

Science

Use supplied measurements, diagrams, safe simulations, or teacher-led observations. Ask learners to separate result from explanation.

Communication

Rewrite a weak email, turn a transcript into action items, or compare two project updates.

Keep activity setup proportional to the learning time. A 15-minute exercise should not require twenty minutes of software installation or data preparation.

10. Lesson review before use

Before using a lesson, read it as if you were the learner. Check whether every required term is explained, whether examples match the level, and whether the assessment tests what was taught.

Quick review checklist

- Do objectives match the source outline?
- Does the lesson fit the requested duration?
- Are examples correct and complete?
- Are important conditions or exceptions preserved?
- Does practice progress in difficulty?
- Do quiz questions match the lesson?
- Are answers and explanations correct?
- Is anything introduced in the quiz that never appeared in the lesson?

A review can result in small edits rather than a complete rewrite. If one example is too hard, change that example without unnecessarily changing the whole lesson.